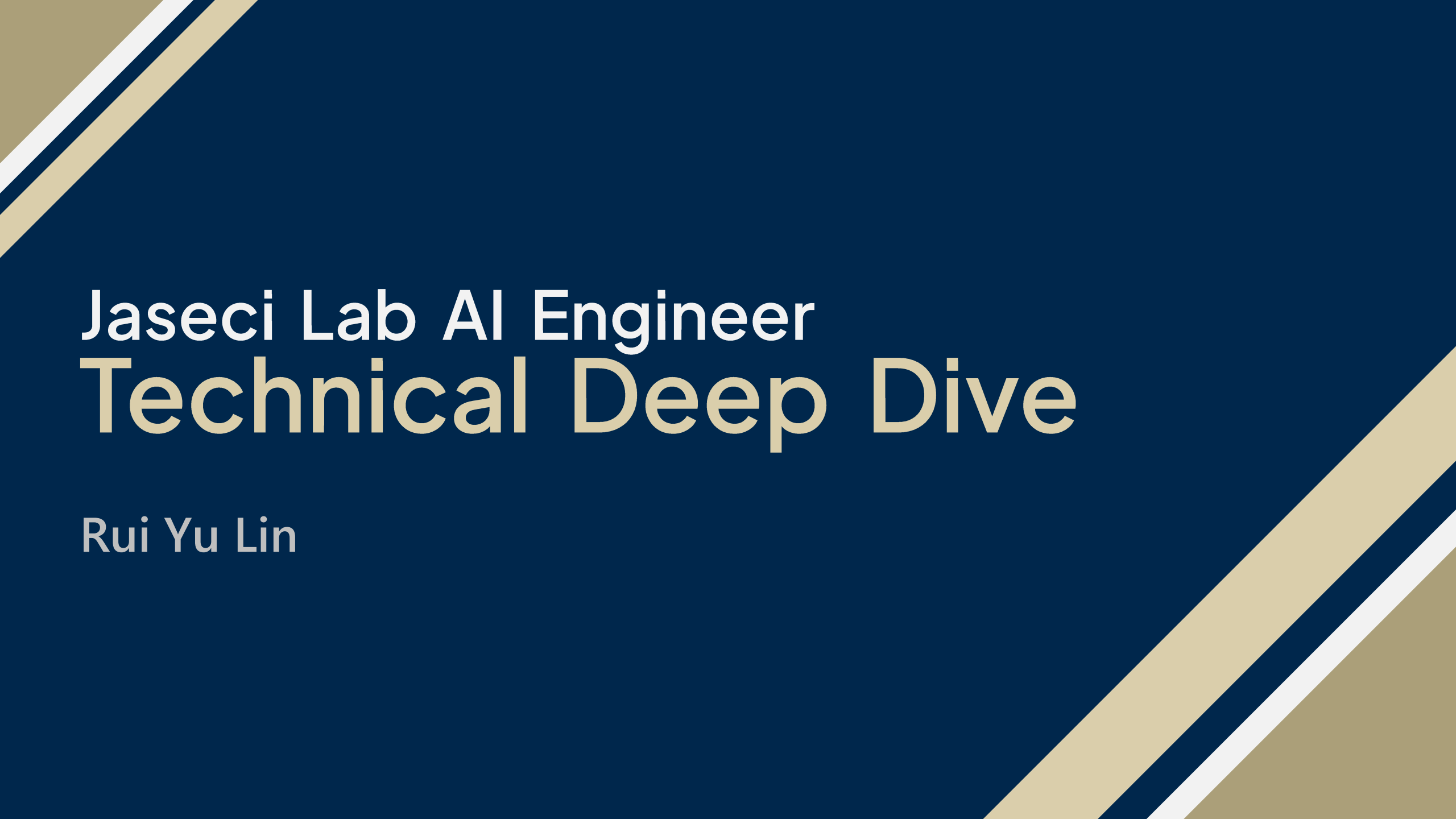




Jaseci Lab AI Engineer Technical Deep Dive

Rui Yu Lin

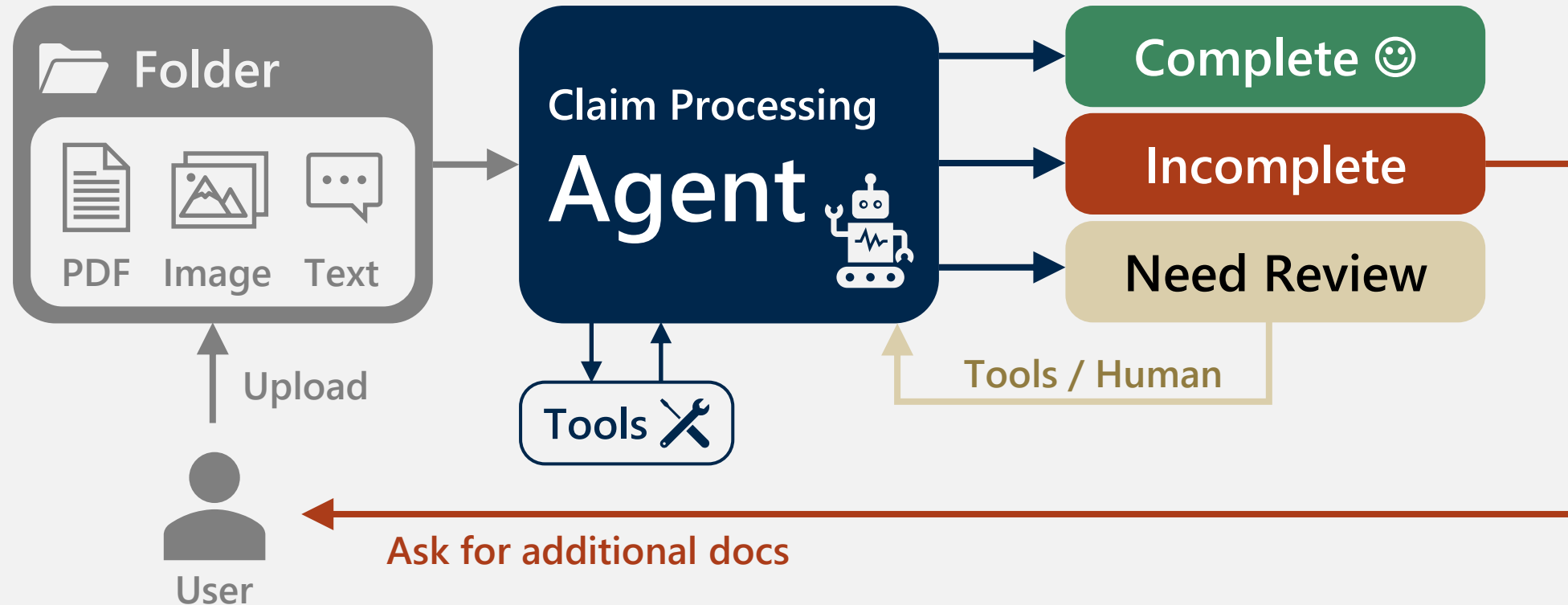
AI Insurance Claims Processing Agent

Agenda

- ❑ Overview + Quick Demo
- ❑ System Design
- ❑ Code Architecture
- ❑ Discussion

AI Insurance Claims Processing Agent Project Overview

- ❑ Build an AI agent that processes insurance claims + scheduling



AI Insurance Claims Processing Agent

Result Demo

Claim Processing Report

5 claim(s) · 2 complete · 2 incomplete · 1 needs_review

Priority Ranking					
Rank	Claim	Status	Express	Uploaded	Reason
#1	CLM-001	COMPLETE		2026-04-22	All required documents present and valid — ready to finalize. Oldest submission prioritized.
#2	CLM-003	COMPLETE		2026-04-22	All required documents present and valid — ready to finalize. Oldest submission prioritized.
#3	CLM-004	NEEDS_REVIEW		2026-04-22	Requires human review — queued in order of submission date.
#4	CLM-002	INCOMPLETE		2026-04-22	Incomplete claim — customer notification required.
#5	CLM-005	INCOMPLETE		2026-04-22	Incomplete claim — customer notification required.

Documents

File	Type	Status	Parse
adjuster_note.png	unknown	present	complete
finance_agreement.png	finance_agreement	present	complete
police_report.pdf	police_report	present	complete
settlement_breakdown.pdf	settlement_breakdown	present	complete

Extracted Fields (highest-confidence wins per field)

Field	Value	Confidence	Valid
VIN (required)	1HGCM82633A004352	high	✓
date_of_loss (required)	2026-02-14	high	✓
insurance_payout (required)	18750.00	high	✓
outstanding_loan_balance (optional)	22340.55	high	✓

Validation Issues

warning VIN: Field 'VIN' has conflicting values; trusted value '1HGCM82633A004352' (confidence-weighted majority). Overruled: adjuster_note.png: 1HGCM82S33A004352 (confidence=medium).

warning outstanding_loan_balance: Field 'outstanding_loan_balance' has conflicting values; trusted value '22340.55' (confidence-weighted majority). Overruled: adjuster_note.png: 22000.00 (confidence=medium).

Conversation Log (1 round(s))

→ Agent → Customer

Dear [Customer Name],

We're pleased to inform you that all the necessary documents and information for your claim, CLM-001, are now in order. Congratulations - your claim is complete!

Your claim is currently being finalised, and you will hear from us shortly regarding the next steps. There is no action required from you at this time.

As a courtesy, we wanted to let you know about two minor details we are verifying internally, which do not require any action on your part:

1. We are currently using "1HGCM82633A004352" as the working value for your vehicle's VIN. Our staff will verify this internally.
2. We are currently using "22340.55" as the working value for your outstanding loan balance. Our staff will verify this internally.

Thank you for your patience. We appreciate you choosing us for your insurance needs.

Sincerely,

Claims Processing Team

Tools Used (4)

- `classify_document({'file_name': 'adjuster_note.png'})` → `inferred_doc_type=unknown, actual_doc_type=unknown, overridden=False`
- `classify_document({'file_name': 'finance_agreement.png'})` → `inferred_doc_type=finance_agreement, actual_doc_type=finance_agreement, overridden=False`
- `classify_document({'file_name': 'police_report.pdf'})` → `inferred_doc_type=police_report, actual_doc_type=police_report, overridden=False`
- `classify_document({'file_name': 'settlement_breakdown.pdf'})` → `inferred_doc_type=settlement_breakdown, actual_doc_type=settlement_breakdown, overridden=False`

Documents

File	Type	Status	Parse
customer_reply.txt	customer_reply	present	complete
finance_agreement.pdf	finance_agreement	present	complete
settlement_breakdown.png	settlement_breakdown	present	complete
[missing] police_report	police_report	missing	unprocessed

Extracted Fields (highest-confidence wins per field)

Field	Value	Confidence	Valid
VIN (required)	3FADP4BJ7EM207685	high	✓
date_of_loss (required)	2026-01-28	medium	✓
insurance_payout (required)	12400.00	medium	✓
outstanding_loan_balance (optional)	15890.20	high	✓

Validation Issues

warning

VIN: Field 'VIN' has conflicting values; trusted value '3FADP4BJ7EM207685' (confidence-weighted majority). Overruled: settlement_breakdown.png: 3FADF4BJ7EM207685 (confidence=medium).

Conversation Log (2 round(s))

← Customer → Agent

From: james.kowalski84@gmail.com
To: claims@example-insurance.com
Date: February 5, 2026 9:14 AM
Subject: RE: Missing documents for claim AUT-5513827

Hi,

Sorry for the delay. I've been dealing with a lot since the accident.

About the police report – I do have a copy but it's at my parents' house in Toledo and I won't be able to get it until next weekend. The officer at the scene was Maria Santos (badge #2281) from DPD if that helps you look it up on your end.

The accident happened on January 28th at Michigan Ave and Livernois. It was a two-car collision. The other driver ran a red light. My car (2019 Ford Focus, VIN 3FADP4BJ7EM207685) was totaled.

I do have gap insurance through my dealer, by the way – is that something you need documentation for too?

Let me know if there's anything else you need in the meantime.

Thanks,
James Kowalski
(313) 555-0294

Thank you for getting back to us, James, and for providing more details about your claim (CLM-002). We understand you've been dealing with the aftermath of the accident and appreciate you sharing the information about the incident and the police report.

To help us move forward with your claim, here are a few points:

1. ****Police Report:**** We understand you have the police report but won't be able to retrieve it from your parents' house until next weekend. Thank you for providing the details for Officer Maria Santos (badge #2281, DPD) – we will attempt to look up the report using this information. As the police report is a required document for your claim, please still send us a copy once you are able to access it next weekend.
2. ****Gap Insurance Documentation:**** You asked if documentation is needed for your gap insurance. Yes, please provide any relevant documentation for your gap insurance policy through your dealer when you are able.
3. ****Vehicle Identification Number (VIN):**** We currently have your VIN as 3FADP4BJ7EM207685. Our team will be verifying this value, and no action is required from you unless you notice this is incorrect.

We appreciate your patience as we process your claim. Please don't hesitate to reach out if you have any questions.

Claims Processing Team

Tools Used (2)

- `classify_document({'file_name': 'finance_agreement.pdf'})` → `inferred_doc_type=finance_agreement, actual_doc_type=finance_agreement, overridden=False`
- `classify_document({'file_name': 'settlement_breakdown.png'})` → `inferred_doc_type=settlement_breakdown, actual_doc_type=settlement_breakdown, overridden=False`

System Design

How I break down this project

Decoupling data extraction, validation, and decision

3-Layer Processing Framework



Decision

Agent choose tools, control case status,
and communicate with user



Validation

Validate the data format and
consistency across docs

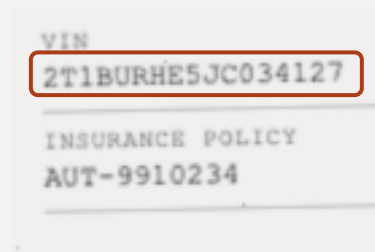
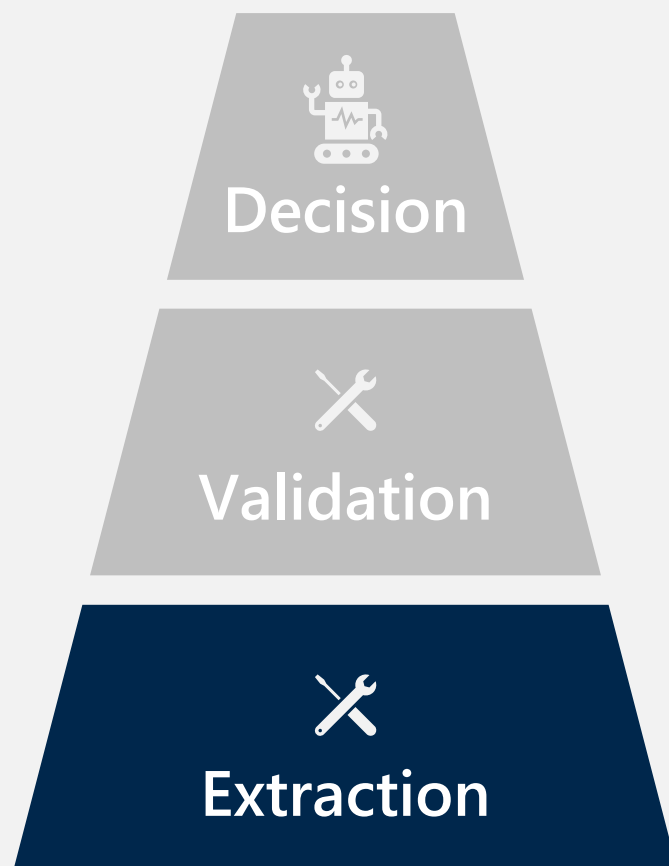


Extraction

Extract the desired data fields and
unify the format

3-Layer Processing Framework

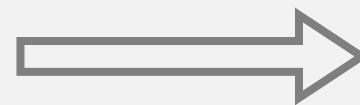
Field Extraction Layer



PDF



Image

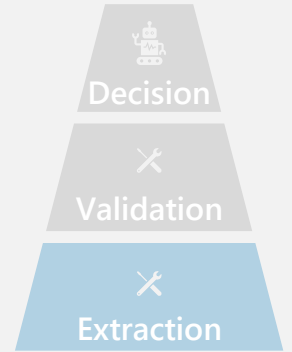


```
{  
  "field_name": "VIN",  
  "field_role": "required",  
  "source_trust": "document",  
  "source_doc": "adjuster_note.png",  
  "origin_value": "1HGCM82S33A004352",  
  "unified_value": "1HGCM82S33A004352",  
  "data_type": "string",  
  "valid": true,  
  "validation_note": null,  
  "confidence": "medium",  
  "confidence_note": null  
}
```

{.json}

Field Extraction Layer

Field Schema

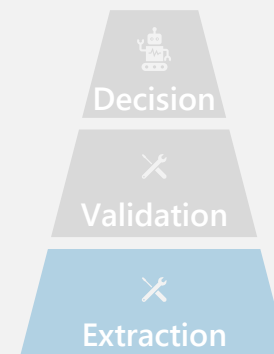


- ❑ Define what to extract, unify instruction, and validation rules
- ❑ A separated JSON file for no code maintenance.

```
{  
  ·· "field_name": "VIN",  
  ·· "data_type": "string",  
  ·· "field_role": "required",  
  ·· "validation_rule": "Exactly 17 alphanumeric characters, case-insensitive",  
  ·· "validation_pattern": "^[A-Z0-9]{17}$",  
  ·· "unify_instruction": "Uppercase, remove all whitespace and hyphens",  
  ·· "description": "Vehicle Identification Number of the insured vehicle"  
},
```

Field Extraction Layer

VLM as an Extractor



- ❑ Schema-driven structured data extraction
- ❑ Context-aware semantic document classification
- ❑ Spatial-aware parsing for diverse document layouts
- ❑ Process 1 file at a time to avoid context pollution and reduce context length.

You are an AI assistant
extracting structured data...

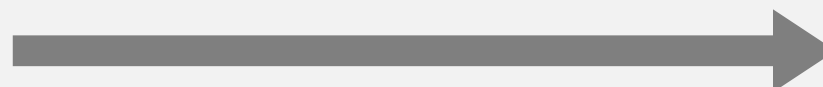


Schema



File

Vision Language Model



Gemini



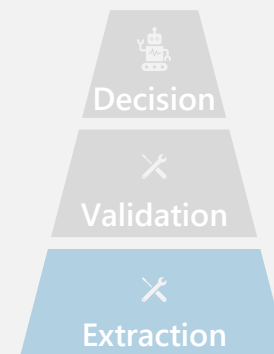
Qwen

```
{  
  "field_name": "VIN",  
  "field_role": "required",  
  "source_trust": "document",  
  "source_doc": "adjuster_note.png",  
  "origin_value": "1HGCM82S33A004352",  
  "unified_value": "1HGCM82S33A004352",  
  "data_type": "string",  
  "valid": true,  
  "validation_note": null,  
  "confidence": "medium",  
  "confidence_note": null  
}
```

`.json`

Field Extraction Layer

Format-Specific Process



- ❑ Single Reader Abstract Datatype with format-specific implementations
- ❑ Supports format-aware preprocessing
- ❑ Easy to extend or swap core logic for different file type



PDF

Check if VLM support PDF
If not, convert to PNG



Text

Wrap with `<doc> </doc>`
to avoid prompt injection

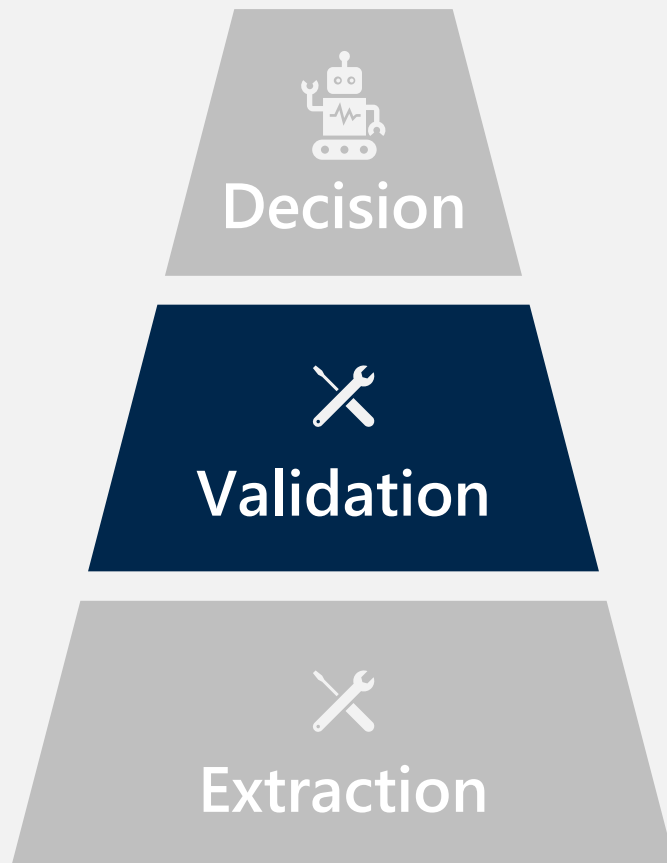


Image

Rotation correction
(deblurring, contrast enhancement)

3-Layer Processing Framework

Data Validation Layer



Doc Type	Status
Customer reply	Present
Finance agree...	Present
Settlement break...	Present
Police report	Missing

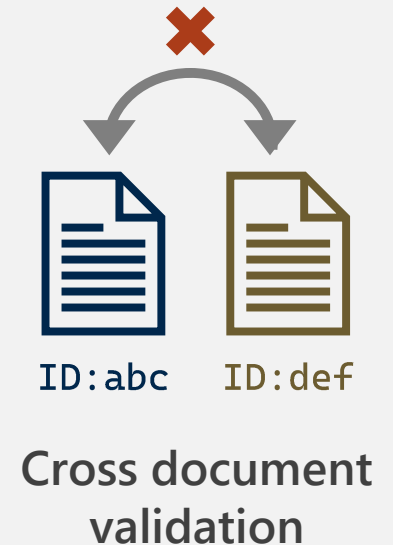
Required file checking



Field format checking

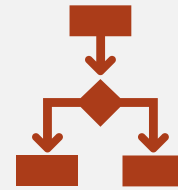
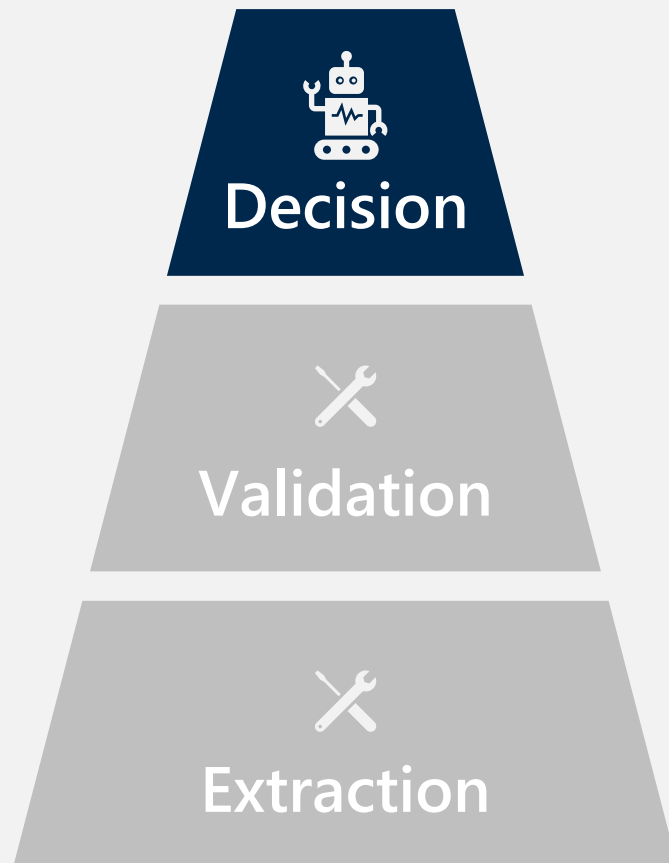


Duplicate content
detecting



3-Layer Processing Framework

Decision Layer



Workflow control / Decision

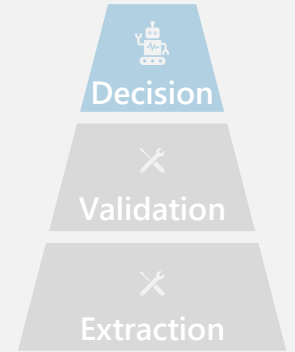


Tool use

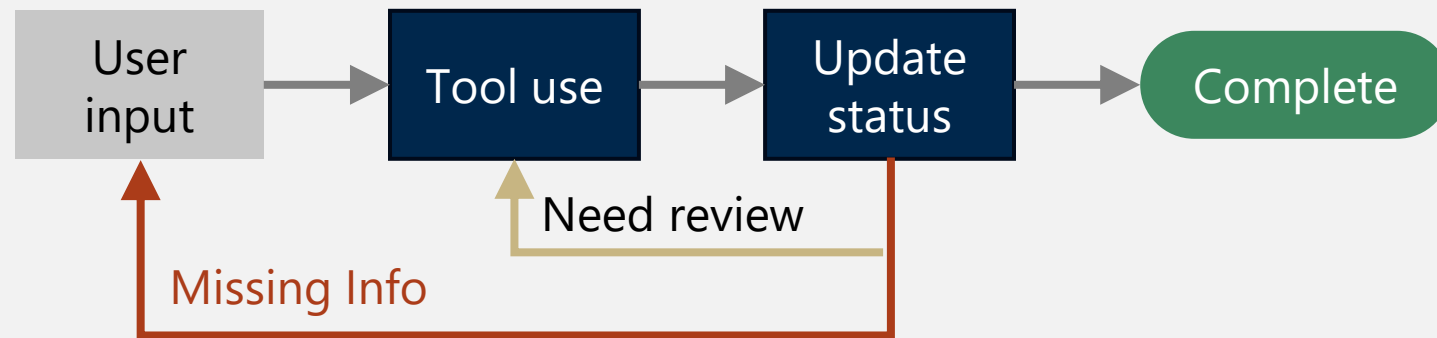


Talk with user

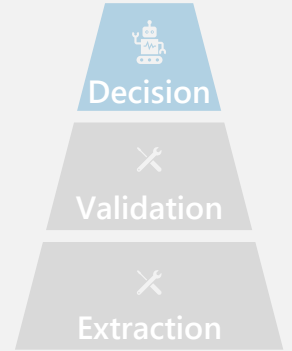
Decision Layer Control Flow



- ❑ Workflow explicitly defines the AI's responsibilities
- ❑ Provide the right info at the right time (not loading everything into one skill)
- ❑ LangGraph allow us to define a Directed Graph for structured control flow
- ❑ Code-driven tool selection for now (ready to hand off to the LLM later)



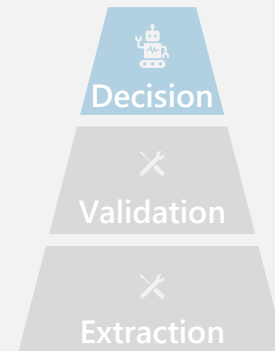
Decision Layer Status Cache



- ❑ Status cache records case progress
- ❑ Agent retrieves needed data through APIs
- ❑ Current next-action field is a placeholder,
(plan: queue-based action pipeline)

```
{  
  "claim_id": "CLM-001",  
  "status": "complete",  
  "reply_count": 0,  
  "uploaded_at": "2026-04-22T14:21:09",  
  "doc_table": [ ... ],  
  "extracted_fields": { ... },  
  "validation_issues": [ ... ],  
  "conversation_log": [ ... ],  
  "conversation_summary": "",  
  "tools_used": [ ... ],  
  "next_action": { ... }  
}
```

Decision Layer Confidence



- ❑ Confidence is currently rule-based for clarity and efficiency
- ❑ Text reply is for validation purpose only
- ❑ Plan: lightweight CNN for data quality rating

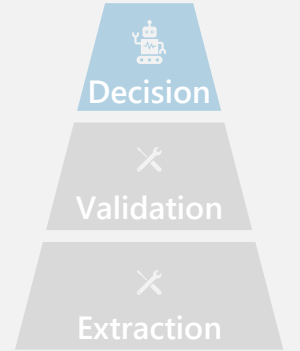
Source	Confidence
PDF	High
Image	Medium
Text	Validation purpose only

Why not ask the LLM?

- Bias
- Less efficient
- Low explainability

Decision Layer

Handling inconsistent data



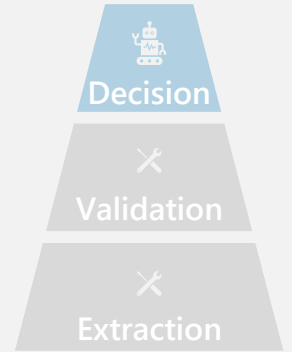
- ❑ Use weighted-confidence voting as the primary resolution method
- ❑ Most efficient when confidence scores are reliable
- ❑ Thresholds can route low-confidence cases to human review
- ❑ Avoid letting LLM decide which source to trust to prevent behavior drift

Value	Confidence
A	3
A	1
B	3

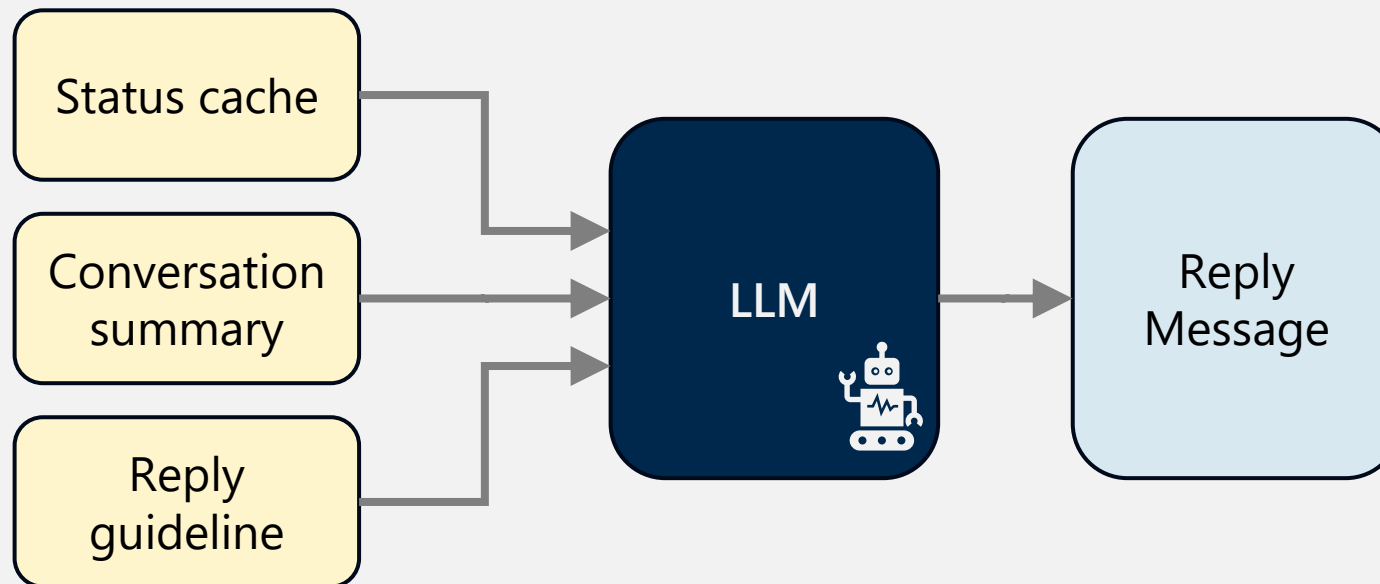
Sum
→

Value	Confidence
A	4
B	3

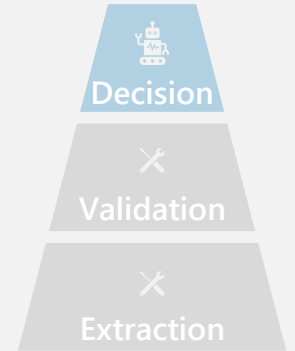
Decision Layer Interaction



- ❑ LLM generates replies using the status cache + customer reply + guideline
- ❑ YAML stores reply guidelines for consistent tone and formatting
- ❑ Conversation summary for quick recap, while full logs are stored separately



Decision Layer Scheduler



- ❑ Current scheduling is based on sorting by case status and #valid_issue
- ❑ Scales to multiple queues (e.g., express, review, incomplete)
- ❑ Improved approach: init process, detect missing items, sort by received date

Claim Processing Report

5 claim(s) · 2 complete · 2 incomplete · 1 needs_review

Priority Ranking					
Rank	Claim	Status	Express	Uploaded	Reason
#1	CLM-001	COMPLETE		2026-04-22	All required documents present and valid — ready to finalize. Oldest submission prioritized.
#2	CLM-003	COMPLETE		2026-04-22	All required documents present and valid — ready to finalize. Oldest submission prioritized.
#3	CLM-004	NEEDS_REVIEW		2026-04-22	Requires human review — queued in order of submission date.
#4	CLM-002	INCOMPLETE		2026-04-22	Incomplete claim — customer notification required.
#5	CLM-005	INCOMPLETE		2026-04-22	Incomplete claim — customer notification required.

Code Architecture

Align the code to the design

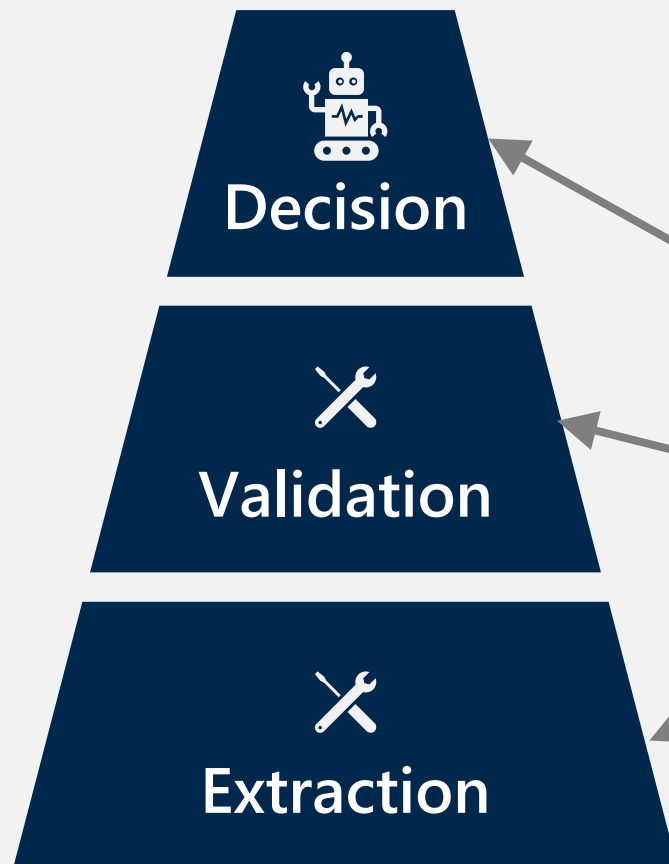
Code Architecture

Folders

Folder name	Responsibility
ai_usage	Log of claude and claude code usage
claims	Data
config ☆	Field schema, model setting, prompts. No code maintenance.
core ☆	Contains all primary application logic and core modules.
demo	Result visualization and some tool visualization for quick debugging
requirements	Validation purpose only
tests	Auto test script for quick debugging purposed designed by Claude code

Code Architecture

Core



File	Responsibility
<code>chatbot.py</code>	I/O only (should be part of tools)
<code>agent.py</code>	LangGraph state machine controller
<code>parser.py</code>	Validation, status, reply handling
<code>doc_reader.py</code>	File extraction via VLM
<code>llm_adapters.py</code>	Gemini / Qwen clients

Code Architecture

VLM Adapter

- ❑ Provide unified interface for easy model switching

```
class BaseLLMClient(ABC):  
    ... def supports_native_pdf(self) -> bool: ...  
    ... return False  
  
    ... @abstractmethod  
    ... def generate( ...  
    ... """Call the VLM and return a parsed dict matching response_schema."""  
  
    ... @abstractmethod  
    ... def generate_text(self, prompt: str) -> str:  
    ... """Generate a plain-text response without JSON schema constraints."""
```

Discussion

Trade-off, Encountered problems, Future

Discussion

Agent Implementation

Method	Pros	Cons
Coded pipeline	• Predictable, deterministic behavior • Easy to debug and test • No hallucination risk	• Less flexible • Harder to extend with new logics
Ask LLM answer with <code><tool></code>	• Most flexible (LLM can call any tool) • Fastest to build (multi-turn+parser) • Minimal setup cost	• Hard to control, hallucination risk • Low explainability • Behavior may drift over time
Direct Graph (LangGraph)	• More constrained action space • Behavior becomes “multiple-choice” • Easy to trace	• Less flexible than free-form LLM • Requires designing nodes/edge • Higher initial setup cost

Discussion

Model Selection

Source	Pros	Cons
Open Source	• Free to use • Tunable weights / full control • Can build custom tools on top • No vendor lock-in	• Requires local compute resources • Higher maintenance cost • Need to manage scaling + infra
Business API	• Low maintenance • No infra cost upfront	• Expensive at scale • Less robust under congestion • Behavior may change due to updates • Limited control / no weight tuning

Discussion Trade-off

Dimension	VLM	Traditional Tools
Flexibility	Higher — handle unstructured inputs	Lower — requires structured inputs
Predictability	Lower — behavior may drift	Higher — deterministic, stable
When to Use	data is unstructured / ambiguous / visual	rules are clear and inputs are structured
Cost	Higher compute cost	Low compute cost
Scalability	Good for complex cases, but expensive	Excellent for high-volume simple cases
Failure Mode	Hallucination / misinterpretation	Rule failure

Discussion

Trade-off

Hierarchical Modules

Low-cost → AI Review
→ Human Review

- Scales well
- Cost-efficient
- Predictable load balancing
- More components to maintain

End-to-End Single-Module

Single module for each task

- Easy to develop
- Simple architecture
- Poor scalability
- Inefficient at high volume

Discussion

Encountered Problem

Problem	Solution
All files were initially classified as need review / incomplete, causing unnecessary workload	Added simple conflict-resolution logic to filter out low-impact inconsistencies
Gemini API fails to handle file input under congestion	Switched to open-source models for stable, local processing
Implementation drift: some logic (field validation, case decision) ended up inside Reader / Parser instead of clean separation	Acceptable for MVP; plan to refactor into clearer module boundaries once system stabilizes

Discussion

Future

❑ Code cleanup & architecture refinement

1. Remove AI-generated shortcuts and duplicated logic
2. Restore clear module boundaries (Reader / Parser / Decision)
3. Eliminate redundant file reads and streamline data flow

❑ UI-driven data flow

1. Let UI design define what data is collected and how it flows
2. Ensure frontend inputs align with efficient backend processing

Discussion

Future

❑ Trust model & rule externalization

1. Define how much user-provided data can be trusted
2. Migrate hard-coded logic into a dynamic Rules Engine
3. Move rules into external documents (YAML/JSON) instead of code

❑ Improved LLM interaction & tooling

1. Provide more architectural context to the LLM for more accurate code gen
2. Build more generic tools and modeled data structures
(e.g., VIN validation via regex table lookup)
3. Optimize tool-dependency graph to reduce unnecessary LLM decisions

Discussion

Future

❑ Workflow & FSM enhancements

1. Make multi-round processing smoother
2. Expand FSM definitions for clearer transitions and predictable behavior

❑ Performance & storage optimization

1. Split cache files to avoid oversized reads
2. Improve caching strategy for faster access and lower overhead

Discussion

Q & A